Java2Compiler - Test Documentation

Running Tests

The Java2Compiler project includes a comprehensive test suite managed by TestRunner.

How to Run Tests

To execute all tests, compile the project and run the following command in your terminal:

Command:

```
java tests.TestRunner
```

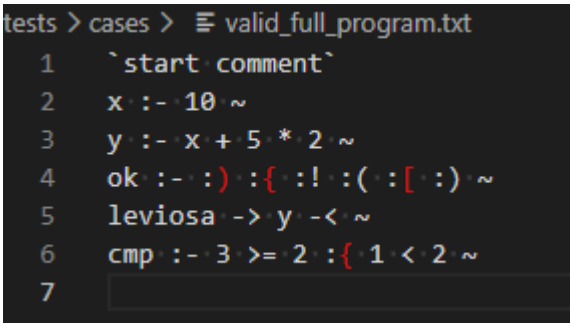
The test runner automatically discovers all test case files in `tests/cases/` and runs them, reporting:

- Valid tests: Should execute without throwing exceptions.
- Invalid tests: Should throw either `LexicalException` (lexer errors) or `ParseException` (parser errors).

Test Cases

Test 1: Full Program (`valid_full_program.txt`)

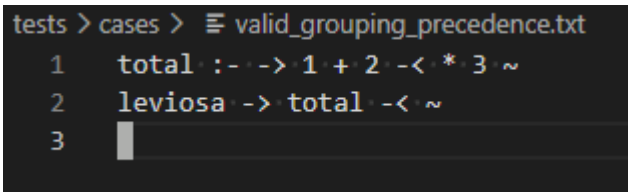
- Type: Valid
- Description: Tests a complete program with comments, assignments, arithmetic operations, boolean expressions, and print statements.
- Test Code:



- Expected Behavior: Program should execute without errors, performing variable assignments and printing output.

Test 2: Grouping and Precedence (`valid_grouping_precedence.txt`)

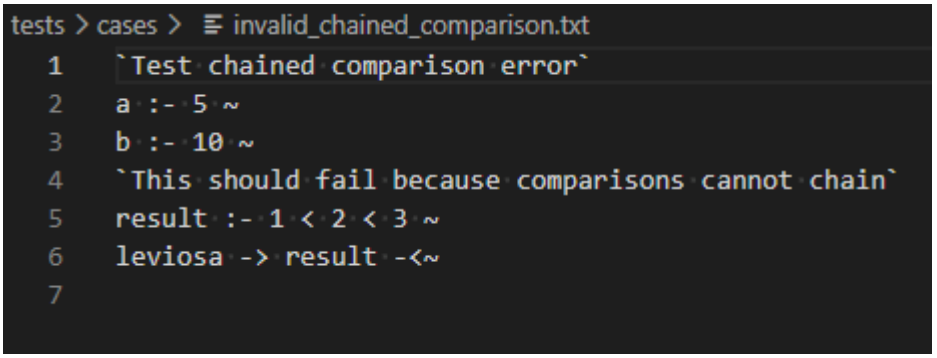
- Type: Valid
- Description: Tests operator precedence and grouping with parentheses.
- Test Code



- Expected Behavior: Should correctly evaluate `(1 + 2) * 3 = 9` and print the result.

Test 3: Chained Comparison (`invalid_chained_comparison.txt`)

- Type: Invalid
- Description: Tests that chained comparisons are rejected (e.g., `1 < 2 < 3`).
- Test Code:



- Expected Behavior: `ParseException` should be thrown - comparisons cannot chain.

Test 4: Expect Expression (`invalid_expect_expression.txt`)

- Type: Invalid
- Description: Tests error handling when an expression is expected but not provided.
- Test Code:

tests > cases > invalid_expect_expression.txt

```
1  `Test missing expression in assignment`
2  a :- 5 ~
3  b :- 10 ~
4  `Missing expression after assignment operator`
5  c :- ~
6
```

- Expected Behavior: `ParseException` should be thrown.

Test 5: Expect Statement (`invalid_expect_statement.txt`)

- Type: Invalid
- Description: Tests error handling when a statement is expected but not provided.
- Test Code:

tests > cases > invalid_expect_statement.txt

```
1  `Test statement with no valid content`
2  x :- 42 ~
3  leviosa -> x -<~
4  `Invalid empty statement`
5  ~
```

- Expected Behavior: `ParseException` should be thrown.

Test 6: Identifier with Digits (`invalid_identifier_digits.txt`)

- Type: Invalid
- Description: Tests rejection of invalid identifiers containing digits.

- Test Code:

tests > cases > invalid_identifier_digits.txt

```
`Test identifier with digits error`
x :- 100 ~
`Identifiers cannot contain digits`
abc1 :- 50 ~
leviosa -> abc1 ~

```

- Expected Behavior: `LexicalException` should be thrown - identifiers cannot contain digits.

Test 7: Missing Assignment Operator (`invalid_missing_assign.txt`)

- Type: Invalid
- Description: Tests error when assignment operator (`:-`) is missing.
- Test Code:

tests > cases > invalid_missing_assign.txt

```
`Test missing assignment operator`
a :- 10 ~
b :- 20 ~
`Missing :- operator`
c 30 ~
leviosa -> a -<~

```

- Expected Behavior: `ParseException` should be thrown.

Test 8: Missing Statement Terminator on Assignment (`invalid_missing_end_assignment.txt`)

- Type: Invalid
- Description: Tests error when statement terminator (`~`) is missing from an assignment.

- Test Code:

tests > cases > invalid_missing_end_assignment.txt

```
1  `Test missing statement terminator`
2  a :- 15 ~
3  b :- 25 ~
4  `Missing ~ terminator on this assignment`
5  c :- 35
6  leviosa -> c -<~
7
```

- Expected Behavior: `ParseException` should be thrown.

Test 9: Missing Statement Terminator on Print (`invalid_missing_end_print.txt`)

- Type: Invalid
- Description: Tests error when statement terminator (`~`) is missing from a print statement.
- Test Code:

tests > cases > invalid_missing_end_print.txt

```
1  `Test missing terminator on print`
2  x :- 99 ~
3  y :- 88 ~
4  `Missing ~ on this print statement`
5  leviosa -> y -<
6  z :- 77 ~
7
```

- Expected Behavior: `ParseException` should be thrown.

Test 10: Missing Right Parenthesis (`invalid_missing_rparen.txt`)

- Type: Invalid
- Description: Tests error when closing parenthesis (`-<`) is missing from a grouped expression.

- Test Code:

tests > cases > invalid_missing_rparen.txt

```
1  `Test missing closing parenthesis`
2  x :- 5 ~
3  y :- 10 ~
4  `Missing closing -< parenthesis`
5  z :- -> x + y ~
6  leviosa -> z -< ~
7
```

- Expected Behavior: `ParseException` should be thrown.

Test 11: Number Overflow (`invalid_number_overflow.txt`)

- Type: Invalid
- Description: Tests rejection of integers that exceed the valid range.
- Test Code:

tests > cases > invalid_number_overflow.txt

```
1  `Test number overflow error`
2  a :- 100 ~
3  b :- 200 ~
4  `Number too large to fit in integer`
5  c :- 999999999999999999999 ~
6  leviosa -> c -<~
7
```

- Expected Behavior: `LexicalException` should be thrown - number overflow.

Test 12: Unexpected Bang (`invalid_unexpected_bang.txt`)

- Type: Invalid
- Description: Tests error when `!` character appears outside of a NOT operator context.
- Test Code:

tests > cases > invalid_unexpected_bang.txt

```
1  `Test unexpected bang character`
2  x :- 42 ~
3  `Invalid ! outside of NOT operator context`
4  !
5  leviosa -> x -< ~
6
```

- Expected Behavior: `LexicalException` should be thrown.

Test 13: Unexpected Character (`invalid_unexpected_char.txt`)

- Type: Invalid
- Description: Tests rejection of invalid characters (e.g., `@`).
- Test Code:

tests > cases > invalid_unexpected_char.txt

```
1  `invalid character`
2  leviosa -> @ -<~
3
```

- Expected Behavior: `LexicalException` should be thrown - unexpected character.

Test 14: Unexpected Colon (`invalid_unexpected_colon.txt`)

- Type: Invalid
- Description: Tests error when `:` appears outside of valid operator contexts.
- Test Code:

tests > cases > invalid_unexpected_colon.txt

```
1  `Test unexpected colon character`
2  x :- 77 ~
3  y :- 88 ~
4  `Invalid : outside operator context`
5  :
6  leviosa -> x -<~
```

- Expected Behavior: `LexicalException` should be thrown.

Test 15: Unterminated Comment (`invalid_unterminated_comment.txt`)

- Type: Invalid
- Description: Tests rejection of comments that are not properly closed.

- Test Code:

tests > cases > invalid_unterminated_comment.txt

```
1  `Test unterminated comment error`
2  x :- 55 ~
3  y :- 66 ~
4  `Unterminated comment with no closing backtick`
5  z :- `this comment is never closed
6  leviosa -> z -<~
```

- Expected Behavior: `LexicalException` should be thrown - unterminated comment.

Test Execution Summary

Test #	Test Name	Type	Expected Result
1	Full Program	Valid	Execute successfully
2	Grouping and Precedence	Valid	Execute successfully
3	Chained Comparison	Invalid	ParseException
4	Expect Expression	Invalid	ParseException
5	Expect Statement	Invalid	ParseException
6	Identifier with Digits	Invalid	LexicalException
7	Missing Assignment	Invalid	ParseException
8	Missing End Assignment	Invalid	ParseException
9	Missing End Print	Invalid	ParseException
10	Missing Right Paren	Invalid	ParseException

11	Number Overflow	Invalid	LexicalException
12	Unexpected Bang	Invalid	LexicalException
13	Unexpected Char	Invalid	LexicalException
14	Unexpected Colon	Invalid	LexicalException
15	Unterminated Comment	Invalid	LexicalException